EXECUTIVE SUMMARY

Shipping Route Efficiency at Nassau Candy Distributor

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Purpose

Nassau Candy ships products from five factories to customers across the United States and Canada, but it has had no clear view of which delivery routes work well and which do not. This project analysed 10,194 order lines (orders placed January 2024 to December 2025, 59 states and provinces) to rank routes, find geographic problem areas and compare shipping methods. The results are also available in a live dashboard.

Important: read the data note first

The shipping times in the source data are not realistic. Orders were placed in 2024 and 2025, but the recorded ship dates fall between 2026 and 2030, giving lead times of about 900 to 1,640 days. The dates appear to be synthetic or shifted.

Every finding below is therefore a comparison between routes, states and shipping methods. None of these figures should be quoted as real delivery times.

Key findings

1. **Problems are local, not regional.** The four customer regions differ by only about 12 days, and the two main factories perform almost identically. The real differences are between individual states (about 217 days between the fastest and slowest states with at least 30 orders).
2. **Tennessee and Indiana are the clearest bottlenecks.** Both have high order volume (183 and 149 orders), about 31% of shipments flagged as delayed, and lead times well above the same-year average (about +45 and +40 days). They are slow when served from either large factory.
3. **Some states are consistently strong.** Virginia (224 orders) and Oregon (124 orders) are faster than average under every measure used, as are smaller states such as Mississippi, Oklahoma and Arkansas. Virginia is fast from both large factories.
4. **The largest markets are not the problem.** California (2,001 orders) and Texas (985 orders) both perform close to the overall average.
5. **Premium shipping showed no speed advantage.** Same Day averaged 1,333 days against 1,314 for Standard Class, while cost per unit and profit margin were nearly identical across all four ship modes.
6. **Order timing distorts the picture.** Orders placed in 2025 are recorded as about 385 days slower than orders placed in 2024. States with more 2025 orders look slower for reasons unrelated to location. The analysis corrects for this, and as a result Washington, New Jersey and several other states that looked slow at first are not confirmed as problem areas.

Performance snapshot

Measure	Value
Order lines analysed	10,194
Average lead time (relative measure, days)	1,320.8
Share of shipments flagged as delayed	21.1%
Routes reliable enough to rank (state level)	96 of 196
Overall gross profit margin	65.9%
Gap between best and worst region	about 12 days
Gap between best and worst state (30+ orders)	about 217 days (166 after correcting for order timing)

Recommended actions, in priority order

1. **Fix the shipping date data.** Confirm how ship dates are recorded and start capturing actual ship and delivery dates. Every decision below should be re-checked against real dates before major spending.
2. **Review Tennessee and Indiana first.** Examine carrier handoffs and local delivery arrangements. Because both main factories are slow into these states, the cause is more likely on the delivery side than at a factory.
3. **Monitor a watch list monthly.** Maryland, Massachusetts, Kentucky and Minnesota are slower than average after correction, and New Mexico and Iowa are slow but low in volume.
4. **Copy what works.** Compare the carrier and handling practices used for Virginia and Oregon with those used for the slow states.
5. **Re-examine premium shipping.** If real dates confirm that faster modes are not faster, review what customers are being promised and charged for.
6. **Use the dashboard routinely.** It ranks routes, maps states and compares ship modes, with filters for dates, regions, states and ship mode.

Where to find more

- **Live dashboard:** https://factorytocustomershippingrouteefficiency.streamlit.app/
- **Full research paper:** Nassau_Candy_Research_Paper.pdf (methods, tables, charts, limitations).
- **Code and data:** the project's GitHub repository (see README).